

CS 229, Fall 2018

Problem Set #3: Deep Learning Unsupervised Learning

Fall 2018

1. [20 points] A Simple Neural Network

Let $X = \{x^{(1)}, \dots, x^{(m)}\}$ be a dataset of m samples with 2 features, i.e. $x^{(i)} \in \mathbb{R}^2$. The samples are classified into 2 categories with labels $y^{(i)} \in \{0, 1\}$. We want to perform binary classification using a simple neural network with a hidden layer of 3 neurons (h_1, h_2, h_3) and one output neuron o .

(a) [5 points] Gradient descent update

Suppose we use the sigmoid function as the activation function for h_1, h_2, h_3 and o . What is the gradient descent update to $w_{1,2}^{[1]}$ assuming we use a learning rate of α ? Your answer should be written in terms of $x^{(i)}$, $o^{(i)}$, $y^{(i)}$, and the weights.

The update rule for the parameter $w_{1,2}^{[1]}$ is given by

$$w_{1,2}^{[1]} := w_{1,2}^{[1]} - \alpha \partial_{w_{1,2}^{[1]}} l,$$

where $l = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - o^{(i)})^2$ is the loss function.

Before doing any calculations, let's see what our network is doing at each step.

$$x \rightarrow h = \sigma(W^{[1]}x + w_0^{[1]})$$

$$h = \sigma(W^{[1]}x + w_0^{[1]}) \rightarrow o = \sigma(w^{[2]T}h + w_0^{[2]})$$

We need to compute $\partial_{w_{1,2}^{[1]}} l$. Applying the chain rule, and based on the flow of the network, we have

$$\partial_{w_{1,2}^{[1]}} l = \frac{\partial l}{\partial o} \frac{\partial o}{\partial h_2} \frac{\partial h_2}{\partial w_{1,2}^{[1]}}$$

Let's go step by step,

$$\frac{\partial l}{\partial o} = \frac{2}{m} \sum_i (o^{(i)} - y^{(i)})$$

$$\frac{\partial o}{\partial h_2} = o^{(i)}(1 - o^{(i)})w_2^{[2]}$$

$$\frac{\partial h_2}{\partial w_{1,2}^{[1]}} = h_2^{(i)}(1 - h_2^{(i)})x_1^{(i)}$$

Therefore

$$\partial_{w_{1,2}^{[1]}} l = w_2^{[2]} \frac{2}{m} \sum_i (o^{(i)} - y^{(i)}) o^{(i)} (1 - o^{(i)}) h_2^{(i)} (1 - h_2^{(i)}) x_1^{(i)},$$

and the update rule is given by,

$$w_{1,2}^{[1]} := w_{1,2}^{[1]} - \alpha \frac{2}{m} w_2^{[2]} \sum_i (o^{(i)} - y^{(i)}) o^{(i)} (1 - o^{(i)}) h_2^{(i)} (1 - h_2^{(i)}) x_1^{(i)}$$

(b) [10 points] Step function weights

Now, suppose instead of using the sigmoid function for the activation function for h_1, h_2, h_3 and o , we instead used the step function $f(x)$ (defined as 1 if $x \geq 0$, else 0). Is it possible to have a set of weights that allow the neural network to classify this dataset with 100% accuracy? If it is possible, provide the weights. If it is not possible, explain your reasoning.

It is possible. In this case, we have 3 neurons, and each one can be treated as an independent linear classifier. The decision boundaries of each classifier must be, in this case, $x_1 = 0.5$, $x_2 = 0.5$, and $x_1 + x_2 = 4$. The output is also a linear classifier, which differentiate points inside and outside the triangle.

To obtain the weights, the equations to verify are:

$$w_{11}^{[1]} x_1 + w_{21}^{[1]} x_2 + w_{01}^{[1]} = 0$$

$$w_{12}^{[1]} x_1 + w_{22}^{[1]} x_2 + w_{02}^{[1]} = 0$$

$$w_{13}^{[1]} x_1 + w_{23}^{[1]} x_2 + w_{03}^{[1]} = 0$$

We need 9 points to solve the system of equations. For the first equation, we could have $\{(0.5, 0), (0.5, 1), (0.5, 2)\}$, and the solution would be $(w_{11}^{[1]}, w_{21}^{[1]}, w_{01}^{[1]}) = (t, 0, -0.5t)$, $t \in \mathbb{R}$. So one solution would be $(-1, 0, 0.5)$.

We could repeat the same process for the other weights and obtain one set of weights that perfectly separate the data.

(c) [10 points] Linear function weights

Let the activation functions for h_1, h_2, h_3 be the linear function $f(x) = x$ and the activation function for o be the same step function as before. Is it possible to have a set of weights that allow the neural network to classify this dataset with 100% accuracy? If yes, provide weights; if no, explain why.

In this case is not possible. If the activation function is linear, then the output of the network will be linear. A linear boundary cannot separate a triangle like the one in the data.

2. [15 points] KL divergence and Maximum Likelihood

The Kullback-Leibler (KL) divergence is a measure of how much one probability distribution is different from a second one. It is a concept that originated in Information Theory, but has made its way into several other fields, including Statistics, Machine Learning, Information Geometry, and many more. In Machine Learning, the KL divergence plays a crucial role, connecting various concepts that might otherwise seem unrelated.

In this problem, we will introduce KL divergence over discrete distributions, practice some simple manipulations, and see its connection to Maximum Likelihood Estimation. The KL divergence between two discrete-valued distributions $P(X)$, $Q(X)$ is defined as

$$D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

For notational convenience, we assume $P(x) > 0, \forall x$.

(a) [5 points] Nonnegativity

Prove the following: $\forall P, Q \quad D_{KL}(P||Q) \geq 0$, and $D_{KL}(P||Q) = 0$ if and only if $P = Q$. (Hint: Use Jensen's inequality).

The KL divergence is given by,

$$D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

This formula can be expressed as,

$$D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} = - \sum_{x \in \mathcal{X}} P(x) \log \frac{Q(x)}{P(x)} = -E_P \left[\log \frac{Q(x)}{P(x)} \right].$$

Now, if we apply Jensen's Inequality,

$$D_{KL}(P||Q) = -E_P \left[\log \frac{Q(x)}{P(x)} \right] \geq -\log E_P \left[\frac{Q(x)}{P(x)} \right] = -\log \sum_{x \in \mathcal{X}} P(x) \frac{Q(x)}{P(x)} = -\log \sum_{x \in \mathcal{X}} Q(x) = -\log 1 = 0$$

Now, we want to show that $D_{KL}(P||Q) = 0 \iff P = Q$. First, if $P = Q$,

$$D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log 1 = 0.$$

Second, if $D_{KL}(P||Q) = 0$,

$$0 = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} = E_P \left[\log \frac{P(x)}{Q(x)} \right] \rightarrow \log \frac{P(x)}{Q(x)} = 0 \rightarrow P(x) = Q(x)$$

(b) [5 points] Chain rule for KL divergence

Prove the chain rule for KL divergence: $D_{KL}(P(X, Y)||Q(X, Y)) = D_{KL}(P(X)||Q(X)) + D_{KL}(P(Y|X)||Q(Y|X))$.

We start with the definition of the divergence of the joint distributions

$$\begin{aligned}
 D_{KL}(P(X, Y) \parallel Q(X, Y)) &= \sum_{x, y} P(x, y) \log \frac{P(x, y)}{Q(x, y)} \\
 &= \sum_{x, y} P(x)P(y|x) \log \frac{P(x)P(y|x)}{Q(x)Q(y|x)} \\
 &= \sum_{x, y} P(x)P(y|x) \left[\log \frac{P(x)}{Q(x)} + \log \frac{P(y|x)}{Q(y|x)} \right] \\
 &= \sum_x P(x) \log \frac{P(x)}{Q(x)} + \sum_x P(x) \sum_y P(y|x) \log \frac{P(y|x)}{Q(y|x)} \\
 &= D_{KL}(P(X) \parallel Q(X)) + D_{KL}(P(Y|X) \parallel Q(Y|X))
 \end{aligned}$$

(c) [5 points] KL and maximum likelihood

Consider a density estimation problem with training set $\{x^{(i)}\}$. Let the empirical distribution be $\hat{P}(x) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{x^{(i)} = x\}$. Prove that finding the maximum likelihood estimate for the parameter θ is equivalent to finding P_θ with minimal KL divergence from $\hat{P}$. I.e., prove:

$$\arg \min_{\theta} D_{KL}(\hat{P} \parallel P_\theta) = \arg \max_{\theta} \sum_{i=1}^m \log P_\theta(x^{(i)})$$

We just need to work on the minimization problem,

$$\begin{aligned}
 \arg \min_{\theta} D_{KL}(\hat{P} \parallel P_\theta) &= \arg \min_{\theta} \sum_{x \in \mathcal{X}} \hat{P}(x) \log \frac{\hat{P}(x)}{P_\theta(x)} \\
 &= \arg \min_{\theta} \sum_{x \in \mathcal{X}} \left[\hat{P}(x) \log \hat{P}(x) - \hat{P}(x) \log P_\theta(x) \right] \\
 &= \arg \max_{\theta} \sum_{x \in \mathcal{X}} \hat{P}(x) \log P_\theta(x) \\
 &= \arg \max_{\theta} \sum_{x \in \mathcal{X}} \left[\frac{1}{m} \sum_{i=1}^m \mathbf{1}\{x^{(i)} = x\} \right] \log P_\theta(x) \\
 &= \frac{1}{m} \arg \max_{\theta} \sum_{i=1}^m \log P_\theta(x^{(i)}).
 \end{aligned}$$

3. [25 points] KL Divergence, Fisher Information, and the Natural Gradient

Let the distribution of a random variable Y parameterized by $\theta \in \mathbb{R}^n$ be $p(y; \theta)$.

(a) [3 points] Score function

The score function is $\nabla_{\theta} \log p(y; \theta)$. Show that the expected value of the score is 0. i.e.

$$\mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] = 0$$

If we develop the expected value,

$$\begin{aligned} \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] &= \int_{-\infty}^{\infty} p(y; \theta) \nabla_{\theta} \log p(y; \theta) dy \\ &= \int_{-\infty}^{\infty} p(y; \theta) \frac{1}{p(y; \theta)} \nabla_{\theta} p(y; \theta) dy \\ &= \int_{-\infty}^{\infty} \nabla_{\theta} p(y; \theta) dy \\ &= \nabla_{\theta} \int_{-\infty}^{\infty} p(y; \theta) dy \\ &= \nabla_{\theta} 1 \\ &= 0. \end{aligned}$$

(b) [2 points] Fisher Information

The Fisher information is defined as the covariance matrix of the score function. Show that the Fisher information can equivalently be given by:

$$\mathcal{I}(\theta) = \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T |_{\theta' = \theta}]$$

The covariance matrix is given by,

$$\text{Cov}(X, X) = E[(X - E[X])(X - E[X])^T].$$

Therefore, taking into account this expression,

$$\begin{aligned} \mathcal{I}(\theta) &= \text{Cov} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[(\nabla_{\theta'} \log p(y; \theta') - E[\nabla_{\theta'} \log p(y; \theta')]) (\nabla_{\theta'} \log p(y; \theta') - E[\nabla_{\theta'} \log p(y; \theta')])^T \right] \\ &= \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T |_{\theta' = \theta}]. \end{aligned}$$

(c) [5 points] Fisher Information (alternate form)

Show that $\mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y; \theta')|_{\theta'=\theta}] = \mathcal{I}(\theta)$.

We start from the Fisher Information formula;

$$\begin{aligned} \mathcal{I}(\theta)_{ij} &= \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T|_{\theta'=\theta}]_{ij} \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{\partial \log p(y; \theta)}{\partial \theta_i} \frac{\partial \log p(y; \theta)}{\partial \theta_j} \right] \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right]. \end{aligned}$$

We want to show that this expression is equal to the one given in the statement of the problem;

$$\begin{aligned} \mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y; \theta')|_{\theta'=\theta}]_{ij} &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} - \frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \int_{-\infty}^{\infty} p(y; \theta) \frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} dy \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \int_{-\infty}^{\infty} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} dy \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \frac{\partial^2}{\partial \theta_i \partial \theta_j} \int_{-\infty}^{\infty} p(y; \theta) dy \\ &= \mathbb{E}_{y \sim p(y; \theta)} \left[\frac{1}{p^2(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] \\ &= \mathcal{I}(\theta). \end{aligned}$$

(d) [5 points] Approximating D_{KL} with Fisher Information

Show that $D_{KL}(p_{\theta} || p_{\theta+d}) \approx \frac{1}{2} d^T \mathcal{I}(\theta) d$ **using Taylor Series expansion.**

We expand until second order the $\log p(y; \bar{\theta})$;

$$\begin{aligned} \log p(y; \bar{\theta}) &\approx \log p(y; \theta) + (\theta - \bar{\theta})^T \nabla_{\theta'} \log p(y; \theta')|_{\theta'=\theta} \\ &\quad + \frac{1}{2} (\theta - \bar{\theta})^T (\nabla_{\theta'}^2 \log p(y; \theta')|_{\theta'=\theta}) (\theta - \bar{\theta}) \\ &= \log p(y; \theta) + d^T \nabla_{\theta'} \log p(y; \theta')|_{\theta'=\theta} \\ &\quad + \frac{1}{2} d^T (\nabla_{\theta'}^2 \log p(y; \theta')|_{\theta'=\theta}) d \end{aligned}$$

If we take expected values;

$$\begin{aligned}
E_{y \sim p(y; \theta)} [\log p(y; \bar{\theta})] &\approx E_{y \sim p(y; \theta)} [\log p(y; \theta)] \\
&\quad + d^T \nabla_{\theta'} E_{y \sim p(y; \theta)} [\log p(y; \theta')] \big|_{\theta' = \theta} \\
&\quad + \frac{1}{2} d^T E_{y \sim p(y; \theta)} [\nabla_{\theta'}^2 \log p(y; \theta') \big|_{\theta' = \theta}] d \\
&= E_{y \sim p(y; \theta)} [\log p(y; \theta)] \\
&\quad + \frac{1}{2} d^T E_{y \sim p(y; \theta)} [\nabla_{\theta'}^2 \log p(y; \theta') \big|_{\theta' = \theta}] d
\end{aligned}$$

Given that;

$$D_{KL}(p(y; \theta) || p(y; \bar{\theta})) = E_{y \sim p(y; \theta)} \left[\log \frac{p(y; \theta)}{p(y; \bar{\theta})} \right]$$

We have;

$$\begin{aligned}
E_{y \sim p(y; \theta)} [\log p(y; \bar{\theta})] - E_{y \sim p(y; \theta)} [\log p(y; \theta)] &\approx \frac{1}{2} d^T E_{y \sim p(y; \theta)} [(\nabla_{\theta'}^2 \log p(y; \theta') \big|_{\theta' = \theta})] d \\
E_{y \sim p(y; \theta)} \left[\log \frac{p(y; \theta)}{p(y; \bar{\theta})} \right] &\approx \frac{1}{2} d^T E_{y \sim p(y; \theta)} [(\nabla_{\theta'}^2 \log p(y; \theta') \big|_{\theta' = \theta})] d = \frac{1}{2} d^T \mathcal{I}(\theta) d
\end{aligned}$$

(e) [8 points] Natural Gradient

Solve the constrained optimization problem for the natural gradient update d^* : $d^* = \arg \max_d l(\theta + d)$ subject to $D_{KL}(p_\theta || p_{\theta+d}) = c$. Construct the Lagrangian, solve for the unscaled natural gradient $\tilde{d}$, and finally the scaled d^* .

The problem can be solved using Lagrangian multipliers. Our Lagrangian is,

$$\mathcal{L}(d, \lambda) = l(\theta + d) - \lambda (D_{KL}(p_\theta || p_{\theta+d}) - c) \approx l(\theta) + d^T \nabla_\theta l(\theta) - \lambda \left[\frac{1}{2} d^T \mathcal{I}(\theta) d - c \right]$$

The solution to the optimization problem is given by,

$$\begin{aligned}
\nabla_d \mathcal{L}(d, \lambda) &= \nabla_\theta l(\theta) - \lambda \mathcal{I}(\theta) d = 0 \\
\nabla_\lambda \mathcal{L}(d, \lambda) &= -\frac{1}{2} d^T \mathcal{I}(\theta) d + c = 0
\end{aligned}$$

From the first equation we have,

$$d = \mathcal{I}^{-1}(\theta) \frac{\nabla_\theta l(\theta)}{\lambda}$$

Substituting in the ∇_λ equation,

$$\begin{aligned}
\frac{1}{2} \frac{(\nabla_\theta l)^T \mathcal{I}^{-1}(\theta) \mathcal{I}(\theta) \mathcal{I}^{-1}(\theta) \nabla_\theta l}{\lambda} + c &= 0 \\
\frac{(\nabla_\theta l)^T \mathcal{I}^{-1}(\theta) \nabla_\theta l}{2\lambda^2} + c &= 0 \\
\lambda &= \sqrt{\frac{1}{2c} (\nabla_\theta l)^T \mathcal{I}(\theta) \nabla_\theta l}
\end{aligned}$$

And therefore,

$$d = \sqrt{\frac{2c}{(\nabla_\theta l)^T \mathcal{I}(\theta) \nabla_\theta l}} \mathcal{I}^{-1}(\theta) \nabla_\theta l(\theta)$$

(f) [2 points] Relation to Newton's Method

Show that the direction of update of Newton's method, and the direction of natural gradient, are exactly the same for Generalized Linear Models (GLMs).

The Newton method is given by,

$$\theta := \theta - H^{-1} \nabla_{\theta} l(\theta)$$

While the natural gradient;

$$\theta := \theta + d = \theta - \mathcal{I}^{-1}(\theta) \frac{\nabla_{\theta} l(\theta)}{\lambda}$$

and we know that

$$\mathcal{I}(\theta) = E_{y \sim p(y; \theta)} [-\nabla_{\theta}^2 \log p(y; \theta)] = E_{y \sim p(y; \theta)} [-\nabla_{\theta}^2 l(\theta)] = -E_{y \sim p(y; \theta)} [H].$$

Thus,

$$\theta := \theta + d = \theta + \mathcal{I}^{-1}(\theta) \frac{\nabla_{\theta} l(\theta)}{\lambda} = \theta - \frac{1}{\lambda} E_{y \sim p(y; \theta)}^{-1} [H] \nabla_{\theta} l(\theta)$$

4. [30 points] Semi-supervised EM

Consider the semi-supervised setting with m unlabeled examples and $\tilde{m}$ labeled examples. The objective is:

$$\ell_{\text{semi-sup}}(\theta) = \ell_{\text{unsup}}(\theta) + \alpha \ell_{\text{sup}}(\theta)$$

(a) [5 points] Convergence

Show that the semi-supervised EM algorithm eventually converges (i.e., show that $\ell_{\text{semi-sup}}(\theta^{(t+1)}) \geq \ell_{\text{semi-sup}}(\theta^{(t)})$).

The objective we want to maximize is,

$$\begin{aligned} \ell_{\text{semi-sup}}(\theta) &= \ell_{\text{unsup}}(\theta) + \alpha \ell_{\text{sup}}(\theta) \\ &= \sum_{i=1}^m \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)}; \theta) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta) \\ &= \sum_{i=1}^m \log \sum_{z^{(i)}} Q_i(z^{(i)}) \frac{p(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta) \\ &= \sum_{i=1}^m \log \mathbb{E}_{Q_i} \left[\frac{p(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right] + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta) \end{aligned}$$

In the E-step we impose that $\log E_{Q_i} \left[\frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} \right] = E_{Q_i} \left[\log \frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} \right]$ (we want Jensen's Inequality to be an equality). In this case our objective is,

$$\begin{aligned} \ell_{\text{semi-sup}}(\theta) &= \ell_{\text{unsup}}(\theta) + \alpha \ell_{\text{sup}}(\theta^t) \\ &= \sum_{i=1}^m \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)}; \theta^t) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \\ &= \sum_{i=1}^m \log \sum_{z^{(i)}} Q_i^t(z^{(i)}) \frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \\ &= \sum_{i=1}^m \log E_{Q_i^t} \left[\frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} \right] + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \\ &= \sum_{i=1}^m E_{Q_i^t} \left[\log \frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} \right] + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \\ &= \sum_{i=1}^m \sum_{z^{(i)}} Q_i^t(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} + \alpha \sum_{i=1}^{\tilde{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \end{aligned}$$

In the M -step we want to maximize this quantity, and therefore

$$\begin{aligned}
\ell_{\text{semi-sup}}(\theta^t) &= \ell_{\text{unsup}}(\theta^t) + \alpha \ell_{\text{sup}}(\theta^t) \\
&= \sum_{i=1}^m \sum_{z^{(i)}} Q_i^t(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^t)}{Q_i^t(z^{(i)})} + \alpha \sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^t) \\
&\leq \sum_{i=1}^m \sum_{z^{(i)}} Q_i^t(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{t+1})}{Q_i^t(z^{(i)})} + \alpha \sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^{t+1}) \\
&\leq \sum_{i=1}^m \log \sum_{z^{(i)}} Q_i^t(z^{(i)}) \frac{p(x^{(i)}, z^{(i)}; \theta^{t+1})}{Q_i^t(z^{(i)})} + \alpha \sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^{t+1}) \\
&= \sum_{i=1}^m \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)}; \theta^{t+1}) + \alpha \sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^{t+1}) \\
&= \ell_{\text{unsup}}(\theta^{t+1}) + \alpha \ell_{\text{sup}}(\theta^{t+1})
\end{aligned}$$

(b) [5 points] Semi-supervised E-Step (GMM)

For the GMM case: Clearly state which latent variables need to be re-estimated and derive the E-step expressions involving x, z, μ, Σ, ϕ .

The latent variables that need to be re-estimated are the $z^{(i)}$'s. In each E-step we set;

$$Q_i(z^{(i)} = j) = p(z^{(i)} = j | x^{(i)}; \mu, \Sigma, \phi)$$

If we take into account that $p(z^{(i)} = j | x^{(i)}; \mu, \Sigma, \phi) = \frac{p(x^{(i)} | z^{(i)} = j; \mu, \Sigma, \phi) p(z^{(i)} = j)}{\sum_{l=1}^k p(x^{(i)} | z^{(i)} = l) p(z^{(i)} = l)}$ and that $p(z^{(i)} = j) = \phi_j$, $x^{(i)} | z^{(i)} = j \sim N(\mu_j, \Sigma_j)$, then

$$\begin{aligned}
Q_i(z^{(i)} = j) &= p(z^{(i)} = j | x^{(i)}; \mu, \Sigma, \phi) = \\
&= \frac{p(x^{(i)} | z^{(i)} = j; \mu, \Sigma, \phi) p(z^{(i)} = j)}{\sum_{l=1}^k p(x^{(i)} | z^{(i)} = l) p(z^{(i)} = l)} = \\
&= \frac{\frac{1}{(2\pi)^{n/2} \Sigma_j^{n/2}} \exp\left(-\frac{1}{2}(x^{(i)} - \mu_j)^T \Sigma_j^{-1} (x^{(i)} - \mu_j)\right) \phi_j}{\sum_{l=1}^k \frac{1}{(2\pi)^{n/2} \Sigma_l^{n/2}} \exp\left(-\frac{1}{2}(x^{(i)} - \mu_l)^T \Sigma_l^{-1} (x^{(i)} - \mu_l)\right) \phi_l}
\end{aligned}$$

(c) [5 points] Semi-supervised M-Step (GMM)

For the GMM case: Clearly state which parameters need to be re-estimated and derive closed form expressions for $\mu^{(t+1)}, \Sigma^{(t+1)}, \phi^{(t+1)}$.

In the M-step, we perform maximization with respect to the parameters of the objective function.

First, we derive with respect the parameter ϕ ,

$$\begin{aligned}
\mathcal{L}(\phi, \mu, \Sigma) &= \sum_i \sum_l Q_i^{(t)}(z^{(i)} = l) \log \frac{p(x^{(i)}, z^{(i)} = l; \phi_l, \mu_l, \Sigma_l)}{Q_i^{(t)}(z^{(i)} = l)} + \alpha \sum_i \log p(\bar{x}^{(i)}, \bar{z}^{(i)}, \theta) \\
&= \sum_i \sum_l Q_i^{(t)}(z^{(i)} = l) \log \frac{p(x^{(i)} | z^{(i)} = l; \phi_l, \mu_l, \Sigma_l) p(z^{(i)} = l)}{Q_i^{(t)}(z^{(i)} = l)} + \alpha \sum_i \log p(\bar{x}^{(i)} | \bar{z}^{(i)}; \theta) p(\bar{z}^{(i)}) \\
&= \sum_i \sum_l Q_i^{(t)}(z^{(i)} = l) (\log N(\mu_l, \Sigma_l) \phi_l - \log Q_i^{(t)}(z^{(i)} = l)) + \alpha \sum_i \log N(\mu_{\bar{z}^{(i)}}, \Sigma_{\bar{z}^{(i)}}) \phi_{\bar{z}^{(i)}} \\
&= \sum_i \sum_l Q_i^{(t)}(z^{(i)} = l) (\log N(\mu_l, \Sigma_l) \phi_l - \log Q_i^{(t)}(z^{(i)} = l)) + \alpha \sum_i \sum_l 1\{\bar{z}^{(i)} = l\} (\log N(\mu_l, \Sigma_l) + \log \phi_l)
\end{aligned}$$

Moreover, we have to take into account the restriction $\sum_{l=1}^k \phi_l = 1$, and take this restriction as a Lagrange multiplier. Therefore,

$$\begin{aligned}
\mathcal{L}(\phi, \mu, \Sigma) &= \sum_i \sum_l Q_i^{(t)}(z^{(i)} = l) \left(\log N(\mu_l, \Sigma_l) + \log \phi_l - \log Q_i^{(t)}(z^{(i)} = l) \right) \\
&\quad + \alpha \sum_i \sum_l 1\{\bar{z}^{(i)} = l\} (\log N(\mu_l, \Sigma_l) + \log \phi_l) \\
&\quad + \lambda \left(\sum_{l=1}^k \phi_l - 1 \right)
\end{aligned}$$

If we take the gradient,

$$\begin{aligned}
\nabla_{\phi_j} \mathcal{L}(\phi, \mu, \Sigma) &= \sum_i Q_i^{(t)}(z^{(i)} = j) \frac{1}{\phi_j} + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \frac{1}{\phi_j} + \lambda = 0 \\
\nabla_{\lambda} \mathcal{L}(\phi, \mu, \Sigma) &= \sum_{l=1}^k \phi_l - 1 = 0
\end{aligned}$$

From the first eq,

$$\phi_j \lambda = - \sum_i Q_i^{(t)}(z^{(i)} = j) - \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \rightarrow \phi_j = -\frac{1}{\lambda} \left(\sum_i Q_i^{(t)}(z^{(i)} = j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \right)$$

And substituting in the second,

$$\begin{aligned}
\sum_{l=1}^k \phi_l &= 1 \\
\sum_i \sum_{l=1}^k Q_i^{(t)}(z^{(i)} = l) + \alpha \sum_I \sum_{l=1}^k 1\{\bar{z}^{(i)} = l\} &= -\lambda \\
\sum_{i=1}^m 1 + \alpha \sum_{i=1}^{\bar{m}} 1 &= -\lambda \\
-(m + \alpha \bar{m}) &= \lambda
\end{aligned}$$

Therefore,

$$\phi_j = \frac{1}{m + \alpha \bar{m}} \left(\sum_i Q^{(t)}(z^{(i)} = j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \right)$$

Now we want to estimate the parameter μ_j ,

$$\begin{aligned} \nabla_{\mu_j} \mathcal{L}(\phi, \mu, \Sigma) &= \sum_i \sum_l Q^{(t)}(z^{(i)} = l) \left(\frac{1}{N(\mu_l, \Sigma_l)} \partial_{\mu_j} N(\mu_l, \Sigma_l) \right) + \alpha \sum_i \sum_l 1\{\bar{z}^{(i)} = l\} \left(\frac{1}{N(\mu_l, \Sigma_l)} \partial_{\mu_j} N(\mu_l, \Sigma_l) \right) \\ &= \sum_i \sum_l Q^{(t)}(z^{(i)} = l) \left[-\frac{1}{2} (\nabla_{\mu_j} \mu_l^\top) \Sigma_l^{-1} (x^{(i)} - \mu_l) - \frac{1}{2} (x^{(i)} - \mu_l)^\top \Sigma_l^{-1} (\nabla_{\mu_j} \mu_l) \right] \\ &\quad + \alpha \sum_i \sum_l 1\{\bar{z}^{(i)} = l\} \left[-\frac{1}{2} (\nabla_{\mu_j} \mu_l^\top) \Sigma_l^{-1} (\bar{x}^{(i)} - \mu_l) - \frac{1}{2} (\bar{x}^{(i)} - \mu_l)^\top \Sigma_l^{-1} (\nabla_{\mu_j} \mu_l) \right] \\ &= \sum_i Q^{(t)}(z^{(i)} = j) \Sigma_j^{-1} (x^{(i)} - \mu_j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \Sigma_j^{-1} (\bar{x}^{(i)} - \mu_j) = 0. \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_i Q^{(t)}(z^{(i)} = j) \Sigma_j^{-1} (x^{(i)} - \mu_j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \Sigma_j^{-1} (\bar{x}^{(i)} - \mu_j) &= 0 \\ \sum_i Q^{(t)}(z^{(i)} = j) (x^{(i)} - \mu_j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} (\bar{x}^{(i)} - \mu_j) &= 0 \\ \sum_i Q^{(t)}(z^{(i)} = j) x^{(i)} + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \bar{x}^{(i)} - \mu_j \left(\sum_i Q^{(t)}(z^{(i)} = j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \right) &= 0 \\ \mu_j &= \frac{\sum_i Q^{(t)}(z^{(i)} = j) x^{(i)} + \alpha \sum_i 1\{\bar{z}^{(i)} = j\} \bar{x}^{(i)}}{\sum_i Q^{(t)}(z^{(i)} = j) + \alpha \sum_i 1\{\bar{z}^{(i)} = j\}} \end{aligned}$$

For the covariance matrix, the computation is similar to that of μ_j . In this case,

$$\nabla_{\Sigma_j} \mathcal{L} = -\frac{1}{2} \Sigma_j^{-1} \left(\sum_{i=1}^m Q^{(t)}(z^{(i)} = j) + \alpha \sum_{i=1}^{\bar{m}} 1\{\bar{z}^{(i)} = j\} \right) + \frac{1}{2} \Sigma_j^{-1} \left(\sum_{i=1}^m Q^{(t)}(z^{(i)} = j) (x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^\top + \alpha \sum_{i=1}^{\bar{m}} 1\{\bar{z}^{(i)} = j\} (\bar{x}^{(i)} - \mu_j)(\bar{x}^{(i)} - \mu_j)^\top \right)$$

Therefore,

$$\Sigma_j = \frac{\sum_{i=1}^m Q^{(t)}(z^{(i)} = j) (x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^\top + \alpha \sum_{i=1}^{\bar{m}} 1\{\bar{z}^{(i)} = j\} (\bar{x}^{(i)} - \mu_j)(\bar{x}^{(i)} - \mu_j)^\top}{\sum_{i=1}^m Q^{(t)}(z^{(i)} = j) + \alpha \sum_{i=1}^{\bar{m}} 1\{\bar{z}^{(i)} = j\}}$$

(d) [5 points] [Coding] Classical (Unsupervised) EM

Implement traditional EM. Submit three scatter plots of the resulting cluster assignments.

Solution in the PS3 jupyter notebook.

(e) [7 points] [Coding] Semi-supervised EM

Implement semi-supervised EM. Submit three scatter plots of the resulting cluster assignments.

Solution in the PS3 jupyter notebook.

(f) [3 points] Comparison

Briefly describe the differences seen in: (i) Number of iterations, (ii) Stability, (iii) Overall quality of assignments.

Solution in the PS3 jupyter notebook.

5. [20 points] K-means for compression

We will use K-means to compress images by reducing the number of colors to $k = 16$.

(a) [15 points] [Coding] K-Means Compression Implementation

Run K-means on ‘peppers-small.tiff’. Apply the centroids to ‘peppers-large.tiff’. Display the new image and compare visually. Include code and the compressed image.

Solution.

□

(b) [5 points] Compression Factor

If we represent the image with these reduced (16) colors, by (approximately) what factor have we compressed the image?

Solution.

□